



Yashraj Singh Pahwa

Roll Number: 1024150095

B. Eng. – Electronics and Computer Engineering

Thapar Institute of Engineering and Technology, Patiala

+91-9899395955

ypahwa_be24@thapar.edu

pahwayashraj@gmail.com

github.com/yashrajpahwa

linkedin.com/in/yashrajpahwa

PROFESSIONAL EXPERIENCE

Data Science Intern

June 2026 – July 2026

Redington

Chennai, Tamil Nadu, India

- Developed components of an Agentic RAG solution for structured databases (PostgreSQL), enabling natural-language querying and insight generation from enterprise data.
- Implemented and evaluated vector-based and vectorless retrieval approaches, comparing retrieval quality, response accuracy, latency, and operational cost using defined performance metrics.
- Explored retrieval optimization techniques for vectorless architectures, including schema-aware search, relationship-based retrieval, and query enhancement methods to improve response relevance.
- Supported deployment and testing of RAG workloads in Amazon Web Services environments, contributing to data pipelines, model integration through bedrock, and system validation.

Project Intern under TIET

January 2025 – May 2025

College's Official Placement Portal

- Developed a platform aimed at centralizing placement data access for recruiters, staff & students.
- Implemented and integrated an OAuth based login allowing for a consistent login experience.
- Delivered a mobile-first user interface ensuring accessibility across devices of all form factors.
- Ensured reliability during testing workloads through a serverless architecture on Google Cloud.

EDUCATION

Degree	Institute / Board	CGPA / Percentage	Year
B. Engineering	Thapar Institute of Engineering & Technology, Patiala <i>Pursuing Electronics and Computer Engineering</i>	8.06 / 10	2028
Senior Secondary	Bluebells School International, New Delhi <i>CBSE Board – PCM with Computer Science</i>	85.2%	2024
Secondary	Bluebells School International, New Delhi <i>CBSE Board – 100% in Mathematics</i>	97.2%	2022

PROJECTS, CONTRIBUTIONS & ACHIEVEMENTS

International Innovation Challenge organized by SAC – ISRO & Manipal University Jaipur ([post](#))

- Ranked **8th among 760 teams** across India among all categories.
- Developed an **AI powered multi-unit edge computing solution** to manage crowds at scale.

MLSC Thapar Timetable — Open-Source Contribution ([timetable](#))

- Improved user experience by enabling local saving of course names, **eliminating the need to repeatedly reference subject codes** by applying DOM manipulation techniques to streamline the timetable interface for users.

Conceptometry ([source](#))

- Developed a **user-centric** and **open-source** Learning Management System (LMS).
- Engineered CI/CD pipelines using Docker, GitHub, and other tools, **streamlining the cloud deployment workflow** and ensuring **minimal human intervention**

SKILLS

- Programming Languages:** C++, C, JavaScript, Python, Golang
- Frameworks & Libraries:** ExpressJS, FastAPI, LangChain, NodeJS, ReactJS, Zustand
- Databases:** ChromaDB, Firebase, MySQL, MongoDB, PostgreSQL, Redis, SQLite
- DevOps & MLOps:** AWS, Cloudflare, CircleCI, Docker, Digital Ocean, LiteLLM, LangFuse, Slurm
- Open Standards:** OpenTelemetry, OAuth, OpenAPI, Model Context Protocol
- Fundamentals:** DBMS, Data Structures & Algorithms, Computer Networking, OOPS, Operating System